



SARS-CoV-2, influenza and RSV monitoring through Belgian wastewaters

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In a nutshell

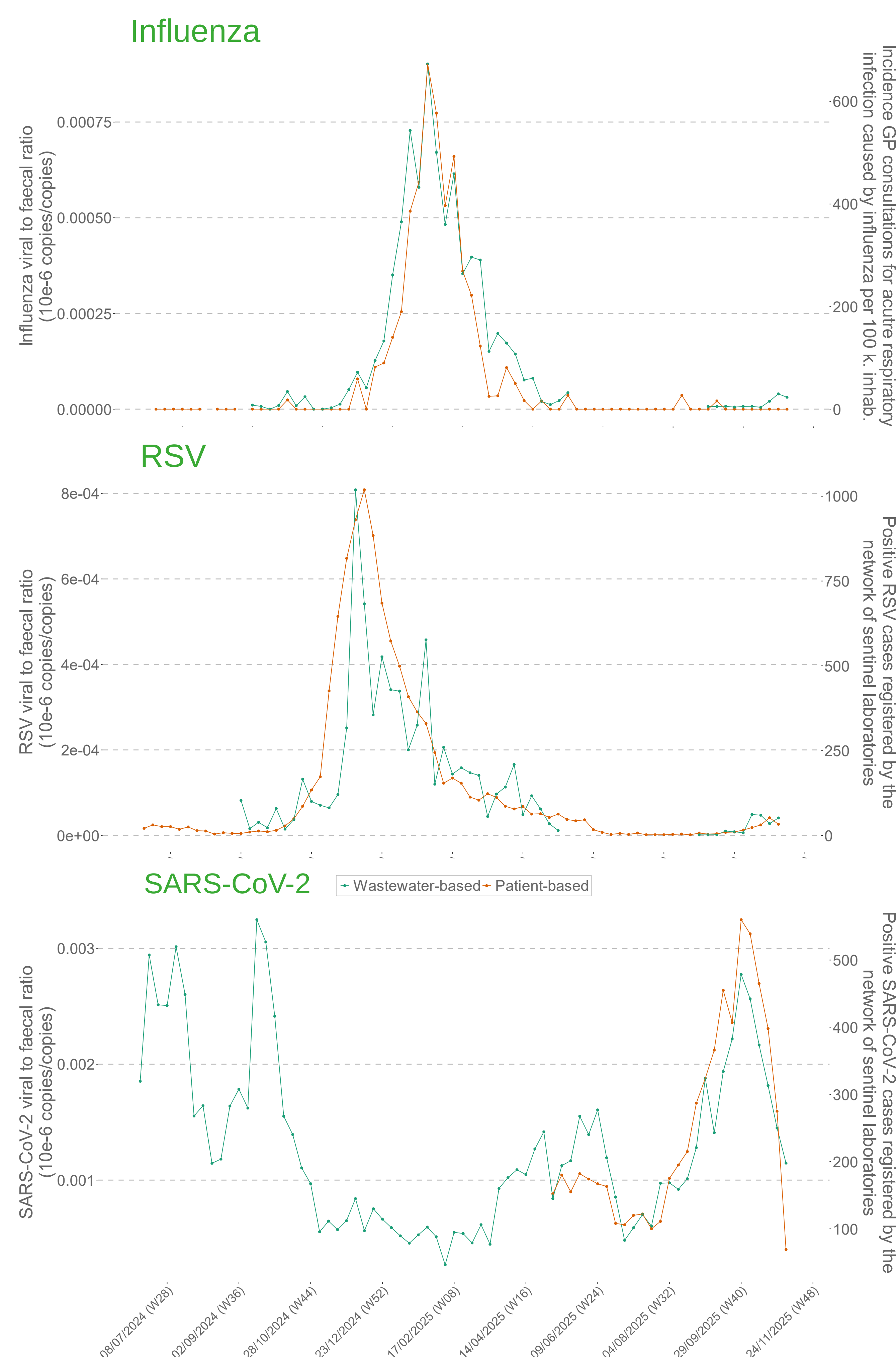
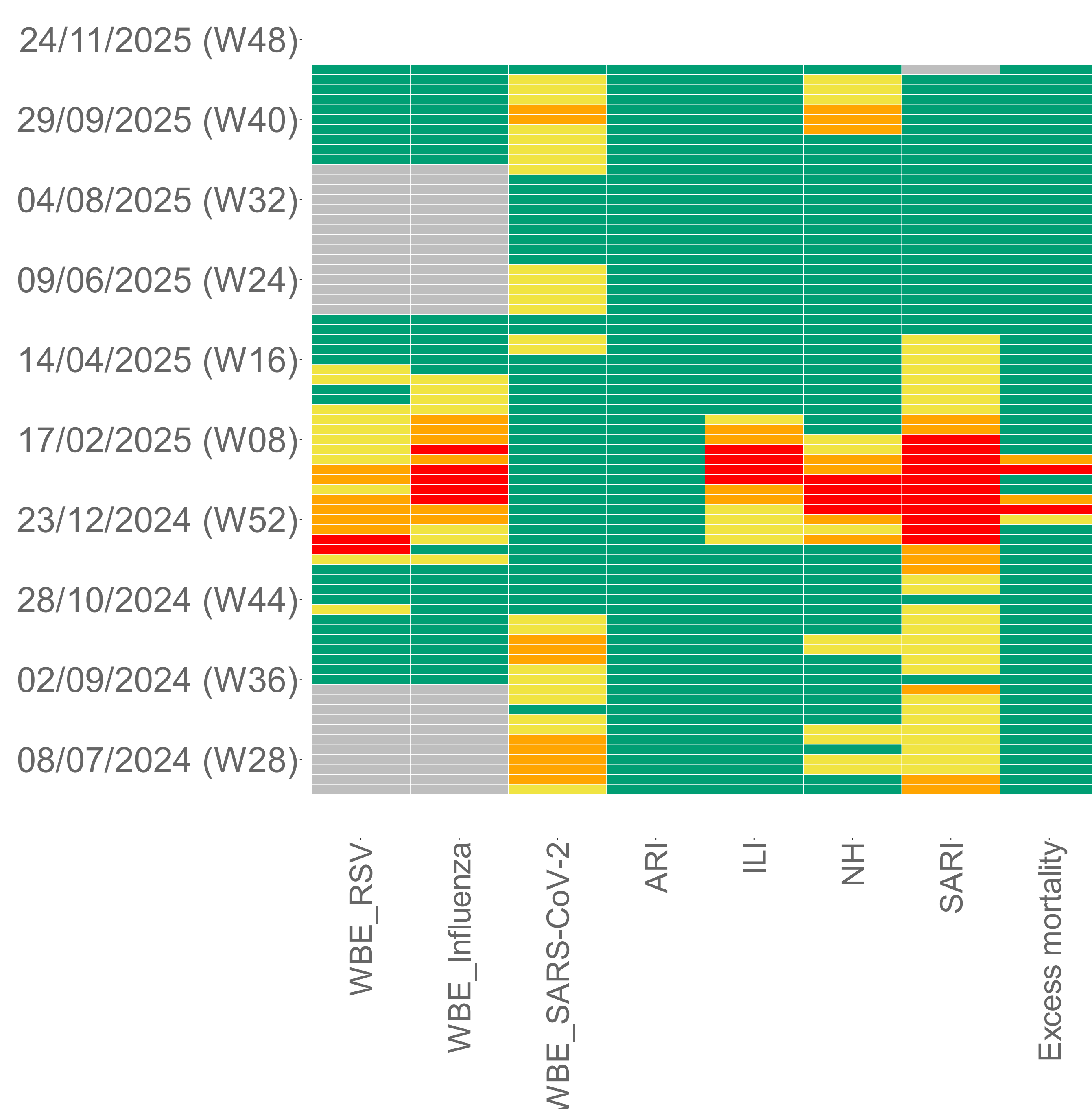
- 38% of the Belgian population monitored throughout 30 wastewater treatment plants
- Surveillance of SARS-CoV-2, influenza, RSV
- Significant correlation with patient-based data
- Added value: Large population covered, short reporting time, no data consolidation

Introduction

Wastewater-based epidemiology enables the development of cost-effective surveillance programs, compared to individual clinical tests for monitoring viruses circulation in a large population. In this work, we present the national wastewater-based epidemiology surveillance used by the public health authority to monitor SARS-CoV-2, RSV and influenza.

Methods

- Ultrafiltration with Centricon or Nanotrap extraction + RT-dPCR multiplex
- Viral ratio computed with the PMMoV faecal marker
- SARS-CoV-2 surveillance with mandatory notification ended in May 2025
- Activity levels reported on heatmap figure: low in green, moderate in yellow, high in orange, and very high in red.
- Spearman correlation computed using R software (version 4.4.2)



REFERENCES

- SARS-CoV-2 Surveillance in Belgian Wastewater, Janssens et al., 2022, MDPI viruses
- Time series modelling for WBE of COVID-19, Bertels et al., 2023, STOTEN

ACKNOWLEDGEMENTS

The contribution of all involved partners is acknowledged: UAntwerp, SPGE, Aquafin, Hydria and Aquiris. TAS team is also acknowledged.

This research was funded by the European Commission, Belgian Federal Government, AVIQ, Vivalis and Department Zorg Vlaanderen.

Results

- High and significant levels of correlation obtained with patient-based data: 0.87 for influenza (n = 47), 0.86 for RSV (n = 47), and 0.84 for SARS (n = 27).
- Different population monitored for RSV through wastewater and sentinel laboratory data. Infants using diapers are not covered by the wastewater samples.
- Added value: large population covered, short reporting time, no data consolidation
- Wastewater data are effectively integrated into the existing surveillance system of respiratory viruses. Integrated reports are weekly communicated to the Risk Assessment Group and transferred in case of need to the Risk Management Group, which then decides on relevant measures to be taken for Public Health.

Weekly report Available At



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